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Acupuncture for biliary colic: a systematic review protocol

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Title Page

Title

Acupuncture for biliary colic: a systematic review protocol

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Acupuncture for biliary colic: a systematic review protocol

ABSTRACT

Introduction Biliary colic (BC) is a severe pain associated with nausea and vomiting, which is the most common symptoms among the gallstone population. This protocol proposes a methodology for conducting a systematic review and meta-analysis that aims to assess the benefits and safety of acupuncture in BC patients.

Methods and analysis Clinical trials will be identified through nine databases from inception to December 2020, using Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, Allied and Complementary Medicine Database (AMED), CINAHL, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Database (CBM), VIP Database and Wanfang Database. Search words will be used for the BC and acupuncture. The analysis would include randomised, controlled, clinical trials of adults with BC that were published in either Chinese or English. The primary outcome is to measure pain relief. Two or three reviewers should be in charge of study selection, data extraction and evaluating the risk of bias. RevMan software (V.5.3) will be used to perform the assessment of the risk of bias and data synthesis.

Ethics and dissemination Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

Trial registration number CRD42020167510

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Strengths and limitations of this study

- This study presents an explicit and replicable methodology for analyzing the effect of biliary colic acupuncture to provide a comprehensive synthesis of the evidence to people with BC, clinicians and policy makers.
- Two or more reviewers will independently perform the study selection, data extraction and assessment of the risk of bias.
- The differences between acupuncture treatments could be a source of heterogeneity between the studies and could cause considerable heterogeneity in this meta-analysis.
- Since different scales may be used in trials, the pooling of analysis of all included studies may not be possible; however, subgroup analyses will be conducted according to different outcomes.

INTRODUCTION

Description of the condition

Biliary colic (BC) is an episodic attack of severe pain for at least 20–30 minutes in the right upper abdominal quadrant, which even radiates to the right-back or shoulder.[1] BC occurs within a couple of hours after eating a meal. It is the most common manifestation of gallstones.[2] BC often accompanying by nausea, vomiting or fever, affects at least one-third of the adult population with gallstones.[3]

When a patient presents with BC, the most critical immediate phase is adequate to control the symptoms, including appropriate analgesia. Non-steroid anti-inflammatory drugs (NSAIDs), opioid drugs, and spasmolytic drugs have been widely used to relieve BC. NSAIDs are usually used as the first line of treatment, such as diclofenac,

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ketoprofen and indomethacin.[4] In patients with active peptic ulcer disease, recent gastrointestinal bleeding, or renal insufficiency, NSAIDs must be used with caution, however. At the same time, NSAIDs can increase the risk of severe side effects.[5]

Description of the intervention

Acupuncture as a non-pharmacological treatment is commonly considered a safe and effective treatment for a variety of conditions that cause pain and discomfort.[6] Acupuncture has been reported to have a clinically relevant, lasting effect on chronic pain.[7] In the meantime, acupuncture has a potential role in controlling acute pain, and it has a better safety profile.[8-10] Acupuncture, as an adjunctive treatment led to the decreased dosage of pain killers.[11-13]

How the intervention might work

Clinically, acupuncture affects the motility of the extrahepatic biliary tract, and improves the clinical symptom of BC.[14,15] But the underlining mechanisms are still unknown. Zhao *et al* found that acupuncture has bi-directional effects on gallbladder pressure and Oddi's sphincter, which is possibly attributable to autonomic reflexes and somatovisceral interactions.[16] Studies have observed the regulatory effect of acupuncture on peripheral and central neurotransmitters to discuss the possible mechanisms for the effect of analgesia, such as stimulating the release of endogenous opioid endorphin.[17,18] Meanwhile, the acupuncture-involving brain regions have been mapped over the past 20 years.[19]

Why it is important to perform this review

Nowadays, the number of acupuncture clinical reports for BC has increased gradually

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in recent years.[15,20] Therefore, pivotal questions has been raised that①whether acupuncture an effective therapy for BC; ②whether acupuncture an effective adjunctive therapy to pharmacological treatment for BC. A lack of systematic review (SR) to address such critical questions. Thus, there is a chance for us to conceive a comprehensive review to determine the efficacy and safety of acupuncture for BC.

Objectives

To assess the benefits and safety of acupuncture in patients with BC.

METHODS

Criteria for including studies in this review

Types of studies

RCTs reported in Chinese or English will be included. Others such as animal research, uncontrolled trials, or case reports will be excluded.

Types of participants

All people presenting with BC according to the definitions and diagnostic work-up in the individual trials.

Types of interventions

The acupuncture style considered must involve the insertion of needles at the acupuncture points. This study will include acupuncture therapy which is used alone or as an add-on to the conventional drug. At the same time, other types of acupuncture points without needle insertion into skin (such as laser, moxibustion or massage), will

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be excluded.

Types of comparator(s)/control

The control group will be considered and classified as following:

1. Acupuncture vs. sham/minimal acupuncture or placebo;

2. Acupuncture vs. conventional drugs/routine care

3. Acupuncture with active treatment vs. active treatment alone.

If the purpose of the study is to compare different forms of acupuncture or acupuncture with other complementary and alternative therapies for the treatment of BC, such studies will be excluded.

Types of outcome measures

Primary outcomes

Quantification of pain relief rated after acupuncture treatment in the trials on quantitative scales, such as visual analogue scale.

Secondary outcomes

1. Number of people with cholelithiasis-related complications.

2. Quality of life.

3. Adverse events.

4. The time needed to obtain pain relief: complete pain relief or any small amount of pain relief (as defined by the different trials).

Search methods for identification of studies

Electronics searches

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The following nine databases including Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Allied and Complementary Medicine Database, CINAHL, China National Knowledge Infrastructure, Chinese Biomedical Literature Database, VIP Database and Wanfang Database will be searched from inception to December 2020.

The terms will be searched as following: biliary colic, biliary disease, biliary pain, biliary complication, acute cholecystitis, cholelithiasis-related complication, acupuncture, acupuncture therapy, manual acupuncture, electro-acupuncture and acupoint. The search strategy for MEDLINE is shown in table 1. We would use the equivalent search words in Chinese databases.

Searching other resources

The following clinical trial registries will be used to retrieved ongoing trials: the NIH clinical registry ClinicalTrials.gov, the International Clinical Trials Registry Platform (ICTRP) and the Chinese clinical registry. We would retrieve the relevant SRs and meta-analyses manually and review it to identify additional studies. Useful but incomplete information would be acquired from the contact trial personnel.

Data collection and analysis

Selection of studies

According to the inclusion criteria, all retrieved studies will be assessed by reviewers (YFZ, YC) in the light of titles and abstracts. Full text of the qualified study will be reviewed if necessary. A third reviewer (LYH) will arbitrate if any disagreement occurs. For excluded studies, the exclusion reason will be listed. The procedure of study selection will be shown in a Preferred Reporting Items for Systematic Reviews

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and Meta-Analyses flow chart (figure 1).

Data extraction and management

Two reviewers (YFZ, YC) will double check and collect information from all qualifying studies and transferred into RevMan software (V.5.3).[21] A predefined data acquisition form will be entering with details: author list and affiliation, publication source, country, interventions, outcomes, adverse effects and so on. The acupuncture intervention details will be elaborated according to STandards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA).[22] A third reviewer (LYH) will arbitrate if any discrepancy noticed. We would obtain the information which is not available by contacting the authors.

Assessment of risk of bias

A systematic review of each study for the bias risk will be done by two independent reviewers (NS, YFZ) by using the Cochrane Handbook for Systematic Reviews of Interventions.[23] Six domains which are the bias of selection, performance, detection, attrition, reporting and other sources will be assessed. For each domain, trials are going to be rated as low risk, high risk, or unclear after evaluation.[24] Involved authors will be contacted if anything unclear. Discrepancies will be arbitrated by a third reviewers (LYH) as well.

Measures of treatment effect

Synthesizing and statistically analyzing of the power data will use RevMan V.5.3. The dichotomous data will be analyzed using a risk ratio with 95% CI. The mean difference or standard mean difference (SMD) with 95% CI will be used to analyze

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the continuous data. If different evaluation tools are used, SMD will be used.

Unit of analysis issues

The analysis will be based on the aggregated outcome data, which is caused by the lack of individual patient data.

Dealing with missing data

Whether the data were intentionally missing or 'randomly' missing can be determined by contacting the corresponding author or author concerned. After the investigator concludes that the data are missing randomly, the available data will be analyzed.[25] If not, it is necessary to apply to missing data to the trial's original investigator of the data or the contact listed in the trial registry. We will impute the missing data with replacement values in the event of no reaction from the authors or contact person, treating them as though they were observed. Missing values will be assumed using the last observation carried forward imputation method, followed by intention-to treat analysis.[26] At the same time, if feasible, we will conduct a sensitivity analyses to address the potential impact of missing data.[27] The potential impact of the effect of missing data on the final findings of the review will be addressed in the discussion.

Assessment of heterogeneity

The forest plot survey will use the $\chi 2$ test with a significance level of $p<0.1$ to investigat the statistical heterogeneity. At the same time, the inconsistency will be quantified by calculating the I^2 test. The fixed-effect model will be used to pool the data. If there are heterogeneity tests in these trials that show little or no statistical heterogeneity ($I^2<50\%$). The random-effects model will be used for heterogeneous

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data (50% <I^2<75%). The meta-analysis will not be conducted in the presence of great heterogeneity.

Assessment of reporting biases

A funnel plot for observing the reporting biases will be generated when more than ten studies are included.[25]

Data synthesis

Data synthesis will be performed through RevMan software (V.5.3). Synthesize and analyze the data according to the level of statistical heterogeneity. The fixed-effects model will be used for the merged data if the detected statistical heterogeneity is small or low, otherwise, a random-effects model will be adopted. When necessary, the possible causes will be analyzed or a subgroup analysis will be conducted. If the heterogeneity is considerable in the included trials, no meta-analysis will be performed.

Subgroup analysis and investigation of heterogeneity

Subgroup analysis will be conducted if data is available. Variations will be considered in the characteristics of the treatments for acupuncture, participants, control types. And there will be subgroups to interpret the heterogeneity.

Sensitivity analysis

The robustness of the main decisions made during the monitoring review process will be conducted through a sensitivity analysis. Several decision-making nodes for sensitivity review need to be considered in the system review process, such as

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methodological flaws, small research and data loss. Sensitivity analysis, as suggested in the Cochrane Handbook, includes two steps: in the first step, all major meta-analysis studies need to be included, and secondly, those studies that are known to be eligible. The results of the sensitivity analysis will be provided in the summary table. As shown by the results of the sensitivity analysis, the risk of bias will be discussed during the review process.

Summary of evidence

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system approach will be used to measure the quality of main outcomes by two reviewers (NS, YFZ).[28] And the results will be present in "Summary of findings" tables. The assessments of evidence quality will be rated "high", "moderate", "low" or "very low". [29,30] Evidence of specific studies will be evaluated based on the risk of bias, inaccuracy, inconsistency, publication bias, indirectness, dose–response relation or effect size.

Patient and public involvement

No patient involved.

Ethics and dissemination

Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. All included studies will be in accordance with the Declaration of Helsinki and current ethical standards. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

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DISCUSSION

There are currently several ongoing RCTs and many others completed of acupuncture for BC. This meta-analysis will provide a relatively convincing conclusion as to whether acupuncture is effective and safe for BC. We assume the findings taken from this study would have important consequences for BC people, physicians and policymakers. If this protocol needs to be amended, we shall provide the date of each amendment and, therefore, state the reasons.

Author Contributors NS, F-RL and R-RS designed the systematic review. NS drafted the protocol，R-RS, F-RL and W-WZ revised the manuscript. NS, Y-FZ, YC, S-RC, JZ, G-XX and L-YH participated in the design of data synthesis and analysis plan. All authors have read and approved the publication of the protocol.

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Competing interests None declared.

Patient consent Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

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Table 1 Search strategy used in MEDLINE database

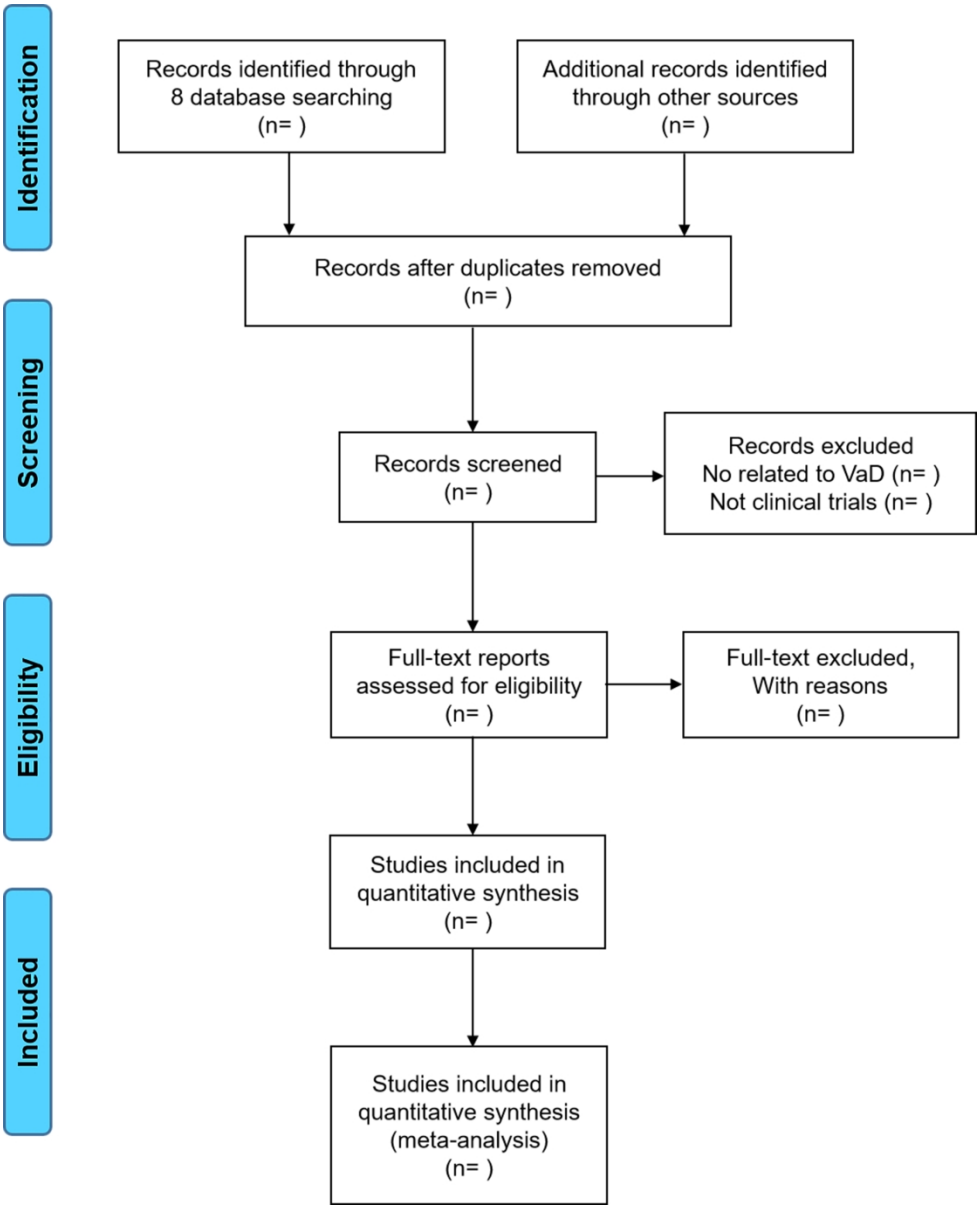
Figure 1 Flow diagram of the study selection process.

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Table 1 Search strategy used in MEDLINE database

NO.	Search items
#1	randomized controlled trial [pt]
#2	controlled clinical trial [pt]
#3	randomized [tiab]
#4	placebo [tiab]
#5	clinical trials [MeSH]
#6	randomly [tiab]
#7	trial [ti]
#8	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
#9	humans [MeSH]
#10	#8 and #9
#11	biliary colic [MeSH]
#12	cholecystitis [MeSH]
#13	(biliary disease or biliary pain or biliary complication or acute cholecystitis or cholelithiasis-related complication): ti,ab
#14	#11 or #12 or #13
#15	acupuncture therapy (MeSH)
#16	(acupuncture or body acupuncture or mamual acupuncture or electroacupuncture or electro-acupuncture or aupoints): ti,ab
#17	#15 or #16
#18	#10 and #14 and #17

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Identification
Screening
Eligibility
Included
Records identified through 8 database searching (n=)
Additional records identified through other sources (n=)
Records after duplicates removed (n=)
Records screened (n=)
Records excluded No related to VaD (n=) Not clinical trials (n=)
Full-text reports assessed for eligibility (n=)
Full-text excluded, With reasons (n=)
Studies included in quantitative synthesis (n=)
Studies included in quantitative synthesis (meta-analysis) (n=)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	
Support:			
Sources	5a	Indicate sources of financial or other support for the review	12
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3-4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-5
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5-6
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be	7

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7-9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	10
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	10
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	11

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Title Page

Title

Acupuncture for biliary colic: a systematic review protocol

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Acupuncture for biliary colic: a systematic review protocol

ABSTRACT

Introduction Biliary colic (BC) is a severe pain associated with nausea and vomiting, which is the most common symptoms among the gallstone population. This protocol proposes a methodology for conducting a systematic review and meta-analysis that aims to assess the benefits and safety of acupuncture in BC patients.

Methods and analysis Clinical trials will be identified through nine databases from inception to December 2020, using Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, Allied and Complementary Medicine Database (AMED), CINAHL, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Database (CBM), VIP Database and Wanfang Database. Search words will be used for the BC and acupuncture. The analysis would include randomised, controlled, clinical trials of adults with BC that were published in either Chinese or English. The primary outcome is to measure pain relief. Two or three reviewers should be in charge of study selection, data extraction and evaluating the risk of bias. RevMan software (V.5.4) will be used to perform the assessment of the risk of bias and data synthesis.

Ethics and dissemination Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

Trial registration number CRD42020167510

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Strengths and limitations of this study

- This study presents an explicit and replicable methodology for analyzing the effect of biliary colic acupuncture to provide a comprehensive synthesis of the evidence to people with BC, clinicians and policy makers.
- Two or more reviewers will independently perform the study selection, data extraction and assessment of the risk of bias.
- The differences between acupuncture treatments could be a source of heterogeneity between the studies and could cause considerable heterogeneity in this meta-analysis.
- Since different scales may be used in trials, the pooling of analysis of all included studies may not be possible; however, subgroup analyses will be conducted according to different outcomes.

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INTRODUCTION

Description of the condition

Biliary colic (BC) is an episodic attack of severe pain for at least 20–30 minutes in the right upper abdominal quadrant, which even radiates to the right-back or shoulder.[1] BC occurs within a couple of hours after eating a meal. It is the most common manifestation of gallstones.[2] BC often accompanying by nausea, vomiting or fever, affects at least one-third of the adult population with gallstones.[3] BC may also occur after cholecystectomy, which is described as postcholecystectomy syndrome (PCS).

When a patient presents with BC, the most critical immediate phase is adequate to control the symptoms, including appropriate analgesia. Non-steroid anti-inflammatory drugs (NSAIDs), opioid drugs, and spasmolytic drugs have been widely used to

relieve BC. NSAIDs are usually used as the first line of treatment, such as diclofenac, ketoprofen and indomethacin.[4] In patients with active peptic ulcer disease, recent gastrointestinal bleeding, or renal insufficiency, NSAIDs must be used with caution, however. At the same time, NSAIDs can increase the risk of severe side effects.[5]

Description of the intervention

Acupuncture as a non-pharmacological treatment is commonly considered a safe and effective treatment for a variety of conditions that cause pain and discomfort.[6] Acupuncture has been reported to have a clinically relevant, lasting effect on chronic pain.[7] In the meantime, acupuncture has a potential role in controlling acute pain, and it has a better safety profile.[8-10] Acupuncture, as an adjunctive treatment led to the decreased dosage of pain killers.[11-13]

How the intervention might work

Clinically, acupuncture affects the motility of the extrahepatic biliary tract, and improves the clinical symptom of BC.[14,15] But the underlining mechanisms are still unknown. Zhao *et al* found that acupuncture has bi-directional effects on gallbladder pressure and Oddi's sphincter, which is possibly attributable to autonomic reflexes and somatovisceral interactions.[16] Studies have observed the regulatory effect of acupuncture on peripheral and central neurotransmitters to discuss the possible mechanisms for the effect of analgesia, such as stimulating the release of endogenous opioid endorphin.[17,18] Meanwhile, the acupuncture-involving brain regions have been mapped over the past 20 years.[19]

Why it is important to perform this review

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Nowadays, the number of acupuncture clinical reports for BC has increased gradually in recent years.[15,20] So far, only one systematic review (SR) published in 2020 resulted that acupuncture plus conventional medicine (CM) with single CM did not show statistical differences in reduction of pain for patients with PCS. [21] However, what is the effect of acupuncture on BC patients without cholecystectomy? Therefore, a lack of SR to address the following pivotal questions: ①whether acupuncture an effective therapy for BC with gallstones; ②whether acupuncture an effective adjunctive therapy to CM for BC with gallstones. Thus, there is a chance for us to conceive a comprehensive review to determine the efficacy and safety of acupuncture for BC.

Objectives

To assess the benefits and safety of acupuncture in patients with BC.

METHODS

Criteria for including studies in this review

Types of studies

RCTs reported in Chinese or English will be included. Others such as animal research, uncontrolled trials, or case reports will be excluded.

Types of participants

All people presenting with BC with gallstones according to the definitions and diagnostic work-up in the individual trials.

Types of interventions

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The acupuncture style considered must involve the insertion of needles at the acupuncture points. This study will include acupuncture therapy which is used alone or as an add-on to the conventional drug. At the same time, other types of acupuncture points without needle insertion into skin (such as laser, moxibustion or massage), will be excluded.

Types of comparator(s)/control

The control group will be considered and classified as following:

1. Acupuncture vs. sham/minimal acupuncture or placebo;

2. Acupuncture vs. conventional drugs/routine care

3. Acupuncture with active treatment vs. active treatment alone.

If the purpose of the study is to compare different forms of acupuncture or acupuncture with other complementary and alternative therapies for the treatment of BC, such studies will be excluded.

Types of outcome measures

Primary outcomes

Quantification of pain relief rated after acupuncture treatment in the trials on quantitative scales, such as visual analogue scale.

Secondary outcomes

1. Number of people with cholelithiasis-related complications.

2. Quality of life.

3. Adverse events.

4. The time needed to obtain pain relief: complete pain relief or any small amount of

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pain relief (as defined by the different trials).

Search methods for identification of studies

Electronics searches

The following nine databases including Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Allied and Complementary Medicine Database, CINAHL, China National Knowledge Infrastructure, Chinese Biomedical Literature Database, VIP Database and Wanfang Database will be searched from inception to December 2020.

The terms will be searched as following: biliary colic, biliary disease, biliary pain, biliary complication, acute cholecystitis, cholelithiasis-related complication, acupuncture, acupuncture therapy, manual acupuncture, electro-acupuncture, acupoint, auricular acupuncture and warm needling. The search strategy for MEDLINE is shown in table 1. We would use the equivalent search words in Chinese databases.

Searching other resources

The following clinical trial registries will be used to retrieved ongoing trials: the NIH clinical registry ClinicalTrials.gov, the International Clinical Trials Registry Platform (ICTRP) and the Chinese clinical registry. We would retrieve the relevant SRs and meta-analyses manually and review it to identify additional studies. Useful but incomplete information would be acquired from the contact trial personnel.

Data collection and analysis

Selection of studies

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According to the inclusion criteria, all retrieved studies will be assessed by reviewers (YFZ, YC) in the light of titles and abstracts. Full text of the qualified study will be reviewed if necessary. A third reviewer (LYH) will arbitrate if any disagreement occurs. For excluded studies, the exclusion reason will be listed. The procedure of study selection will be shown in a Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart (figure 1).

Data extraction and management

Two reviewers (YFZ, YC) will double check and collect information from all qualifying studies and transferred into RevMan software (V.5.4).[22] A predefined data acquisition form will be entering with details: author list and affiliation, publication source, country, interventions, outcomes, adverse effects and so on. The acupuncture intervention details (such as the type of acupuncture, number of needle insertions, depths of insertion, selection of points, de-qi, needle stimulation) will be elaborated according to STandards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA).[23] A third reviewer (LYH) will arbitrate if any discrepancy noticed. We would obtain the information which is not available by contacting the authors.

Assessment of risk of bias

A systematic review of each study for the bias risk will be done by two independent reviewers (NS, YFZ) by using the Cochrane Handbook for Systematic Reviews of Interventions. Six domains which are the bias of selection, performance, detection, attrition, reporting and other sources will be assessed. For each domain, trials are going to be rated as low risk, high risk, or unclear after evaluation.[24] Involved authors will be contacted if anything unclear. Discrepancies will be arbitrated by a third

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reviewers (LYH) as well.

Measures of treatment effect

Synthesizing and statistically analyzing of the power data will use RevMan V.5.4. The dichotomous data will be analyzed using a risk ratio with 95% CI. The mean difference or standard mean difference (SMD) with 95% CI will be used to analyze the continuous data. If different evaluation tools are used, SMD will be used.

Unit of analysis issues

The analysis will be based on the aggregated outcome data, which is caused by the lack of individual patient data.

Dealing with missing data

Whether the data were intentionally missing or 'randomly' missing can be determined by contacting the corresponding author or author concerned. After the investigator concludes that the data are missing randomly, the available data will be analyzed.[25] If not, it is necessary to apply to missing data to the trial's original investigator of the data or the contact listed in the trial registry. In the event of no reaction from the authors or contact person, we will apply the following strategies to address missing data:

1. In the included studies, if intention to-treat (ITT) analyses were performed,[26] we will use ITT data as the first choice, not missing data.

2. For continuous missing outcome data, if the medians, p values or confidence intervals were reported. we will try to re-calculate mean difference and standard deviation values as the first choice.

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At the same time, if feasible, we will conduct a sensitivity analyses to address the potential impact of missing data.[27] The potential impact of the effect of missing data on the final findings of the review will be addressed in the discussion.

Assessment of heterogeneity

Forest plots will be used to present effect sizes and corresponding 95% confidence limits for each study. At the same time, the inconsistency will be quantified by calculating the I^2 test.

Assessment of reporting biases

A funnel plot for observing the reporting biases will be generated when more than ten studies are included.[25] Potential reporting biases will be further assessed with Egger's test.

Data synthesis

Data synthesis will be performed through RevMan software (V.5.4). The random-effects model will be used to pool the data. When necessary, the possible causes of the large degree of heterogeneity will be analyzed or a subgroup analysis will be conducted. If the heterogeneity is considerable in the included trials, no meta-analysis will be performed.

Subgroup analysis and investigation of heterogeneity

Subgroup analysis will be conducted if data is available. Variations will be considered in the characteristics of the treatments for acupuncture, participants, control types. And there will be subgroups to interpret the heterogeneity.

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Sensitivity analysis

The robustness of the main decisions made during the monitoring review process will be conducted through a sensitivity analysis. Several decision-making nodes for sensitivity review need to be considered in the system review process, such as methodological flaws, small research and data loss. Sensitivity analysis, as suggested in the Cochrane Handbook, includes two steps: in the first step, all major meta-analysis studies need to be included, and secondly, those studies that are known to be eligible. The results of the sensitivity analysis will be provided in the summary table. As shown by the results of the sensitivity analysis, the risk of bias will be discussed during the review process.

Other analysis

The use of meta-regression analysis will be considered to explore the potential heterogeneous resources.

Summary of evidence

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system approach will be used to measure the quality of main outcomes by two reviewers (NS, YFZ).[28] And the results will be present in “Summary of findings” tables. The assessments of evidence quality will be rated “high”, “moderate”, “low” or “very low”. [29,30] Evidence of specific studies will be evaluated based on the risk of bias, inaccuracy, inconsistency, publication bias, indirectness, dose–response relation or effect size.

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Patient and public involvement

No patient involved.

Ethics and dissemination

Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. All included studies will be in accordance with the Declaration of Helsinki and current ethical standards. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

DISCUSSION

There are currently several ongoing RCTs and many others completed of acupuncture for BC. This meta-analysis will provide a relatively convincing conclusion as to whether acupuncture is effective and safe for BC. We assume the findings taken from this study would have important consequences for BC people, physicians and policymakers. If this protocol needs to be amended, we shall provide the date of each amendment and, therefore, state the reasons.

Author Contributors NS, F-RL and R-RS designed the systematic review. NS drafted the protocol，R-RS, F-RL and W-WZ revised the manuscript. NS, Y-FZ, YC, S-RC, JZ, G-XX and L-YH participated in the design of data synthesis and analysis plan. All authors have read and approved the publication of the protocol.

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Competing interests None declared.

Patient consent Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

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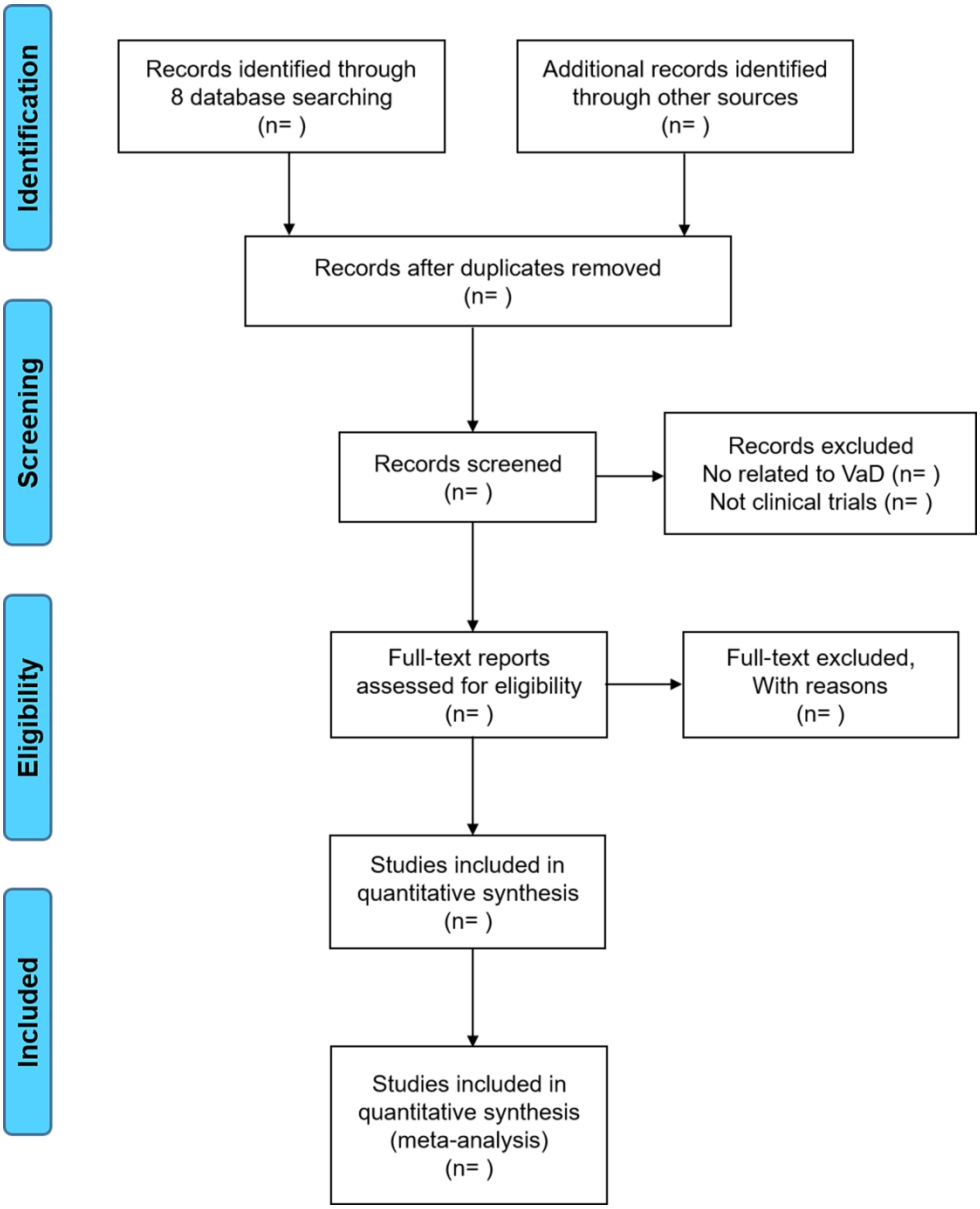
Table 1 Search strategy used in MEDLINE database

Figure 1 Flow diagram of the study selection process.

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Table 1 Search strategy used in MEDLINE database	
NO.	**Search items**
#1	randomized controlled trial [pt]
#2	controlled clinical trial [pt]
#3	randomized [tiab]
#4	placebo [tiab]
#5	clinical trials [MeSH]
#6	randomly [tiab]
#7	trial [ti]
#8	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
#9	humans [MeSH]
#10	#8 and #9
#11	biliary colic [MeSH]
#12	cholecystitis [MeSH]
#13	(biliary disease or biliary pain or biliary complication or acute cholecystitis or cholelithiasis-related complication): ti,ab
#14	#11 or #12 or #13
#15	acupuncture therapy (MeSH)
#16	acupuncture (MeSH)
#17	(acupuncture or body acupuncture or manual acupuncture or electroacupuncture or electro-acupuncture or acupoint or auricular acupuncture or warm needling): ti,ab
#18	#15 or #16 or #17
#19	#10 and #14 and #18

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Identification
Screening
Eligibility
Included
Records identified through 8 database searching (n=)
Additional records identified through other sources (n=)
Records after duplicates removed (n=)
Records screened (n=)
Records excluded No related to VaD (n=) Not clinical trials (n=)
Full-text reports assessed for eligibility (n=)
Full-text excluded, With reasons (n=)
Studies included in quantitative synthesis (n=)
Studies included in quantitative synthesis (meta-analysis) (n=)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	
Support:			
Sources	5a	Indicate sources of financial or other support for the review	12
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3-4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-5
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5-6
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be	7

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		repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7-9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	10
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	10
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	11

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Title Page

Title

Acupuncture for biliary colic: a systematic review protocol

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Acupuncture for biliary colic: a systematic review protocol

ABSTRACT

Introduction Biliary colic (BC) is a severe pain associated with nausea and vomiting, which is the most common symptom among the gallstone population. This protocol proposes a methodology for conducting a systematic review and meta-analysis that aims to assess the benefits and safety of acupuncture in BC patients.

Methods and analysis Clinical trials will be identified through nine databases from inception to December 2020, using Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, Allied and Complementary Medicine Database (AMED), CINAHL, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Database (CBM), VIP Database and Wanfang Database. Search words will be used for the BC and acupuncture. The analysis would include randomised, controlled, clinical trials of adults with BC that were published in either Chinese or English. The primary outcome is to measure pain relief. Two or three reviewers should be in charge of study selection, data extraction and evaluating the risk of bias. RevMan software (V.5.4) will be used to perform the assessment of the risk of bias and data synthesis.

Ethics and dissemination Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

Trial registration number CRD42020167510

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Strengths and limitations of this study

- This study presents an explicit and replicable methodology for analyzing the effect of acupuncture in order to provide a comprehensive synthesis of the evidence for patients with BC, clinicians and policy makers.
- Two or more reviewers will independently perform the study selection, data extraction and assessment of the risk of bias.
- The differences between acupuncture treatments could be a source of heterogeneity between the studies and could cause considerable heterogeneity in this meta-analysis.
- Since different scales may be used in trials, the pooling of analysis of all included studies may not be possible; however, subgroup analyses will be conducted according to different outcomes.

INTRODUCTION

Description of the condition

Biliary colic (BC) is an episodic attack of severe pain for at least 20–30 minutes in the right upper abdominal quadrant, which even radiates to the right-back or shoulder.[1] BC occurs within a couple of hours after eating a meal. It is the most common manifestation of gallstones.[2] BC often accompanied by nausea, vomiting or fever, affects at least one-third of the adult population with gallstones.[3] BC may also occur after cholecystectomy, which is described as postcholecystectomy syndrome (PCS).

When a patient presents with BC, the most critical immediate phase is adequate to control the symptoms, including appropriate analgesia. Non-steroid anti-inflammatory drugs (NSAIDs), opioid drugs, and spasmolytic drugs have been widely used to

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relieve BC. NSAIDs are usually used as the first line of treatment, such as diclofenac, ketoprofen and indomethacin.[4] In patients with active peptic ulcer disease, recent gastrointestinal bleeding, or renal insufficiency, NSAIDs must be used with caution, however. At the same time, NSAIDs can increase the risk of severe side effects.[5]

Description of the intervention

Acupuncture as a non-pharmacological treatment is commonly considered a safe and effective treatment for a variety of conditions that cause pain and discomfort.[6] Acupuncture has been reported to have a clinically relevant, lasting effect on chronic pain.[7] In the meantime, acupuncture has a potential role in controlling acute pain, and it has a better safety profile.[8-10] Acupuncture, as an adjunctive treatment, led to the decreased dosage of pain killers.[11-13]

How the intervention might work

Clinically, acupuncture affects the motility of the extrahepatic biliary tract, and improves the clinical symptom of BC.[14,15] But the underlining mechanisms are still unknown. Zhao *et al.* found that acupuncture has bi-directional effects on gallbladder pressure and Oddi's sphincter, which is possibly attributable to autonomic reflexes and somatovisceral interactions.[16] Studies have observed the regulatory effect of acupuncture on peripheral and central neurotransmitters to discuss the possible mechanisms for the effect of analgesia, such as stimulating the release of endogenous opioid endorphin.[17,18] Meanwhile, the acupuncture-involving brain regions have been mapped over the past 20 years.[19]

Why it is important to perform this review

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In recent years, the number of clinical reports on acupuncture treatment for BC has increased gradually.[15,20] So far, only one systematic review (SR) published in 2020 resulted that there is no statistical difference between acupuncture plus conventional medicine (CM) and single CM in reducing pain for patients with PCS. [21] In this case, a lack of SR to address the following pivotal questions: ①is acupuncture effective in treating biliary colic with gallstones; ②whether acupuncture an effective adjunctive therapy to CM for BC with gallstones. Thus, there is a chance for us to conceive a comprehensive review to determine the efficacy and safety of acupuncture for BC.

Objectives

To assess the benefits and safety of acupuncture in patients with BC.

METHODS

Criteria for including studies in this review

Types of studies

RCTs reported in Chinese or English will be included. Others such as animal research, uncontrolled trials, or case reports will be excluded.

Types of participants

All people who are presenting with BC with gallstones according to the definitions and diagnostic work-up in the individual trials.

Types of interventions

The acupuncture style considered must involve the insertion of needles at the

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acupuncture points. This study will include acupuncture therapy which is used alone or as an add-on to the conventional drug. At the same time, other types of acupuncture points without needle insertion into skin (such as laser, moxibustion or massage), will be excluded.

Types of comparator(s)/control

The control group will be considered and classified as following:

1. Acupuncture vs sham/minimal acupuncture or placebo;

2. Acupuncture vs conventional drugs/routine care

3. Acupuncture with active treatment vs active treatment alone.

If the purpose of the study is to compare different forms of acupuncture or acupuncture with other complementary and alternative therapies for the treatment of BC, such studies will be excluded.

Types of outcome measures

Primary outcomes

Quantification of pain relief rated after acupuncture treatment in the trials on quantitative scales, such as visual analogue scale.

Secondary outcomes

1. Number of people with cholelithiasis-related complications.

2. Quality of life.

3. Adverse events.

4. The time needed to obtain pain relief: complete pain relief or any small amount of pain relief (as defined by the different trials).

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Search methods for identification of studies

Electronics searches

The following nine databases including Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Allied and Complementary Medicine Database, CINAHL, China National Knowledge Infrastructure, Chinese Biomedical Literature Database, VIP Database and Wanfang Database will be searched from inception to December 2020.

The terms will be searched as following: biliary colic, biliary disease, biliary pain, biliary complication, acute cholecystitis, cholelithiasis-related complication, acupuncture, acupuncture therapy, manual acupuncture, electro-acupuncture, acupoint, auricular acupuncture and warm needling. The search strategy for MEDLINE is shown in table 1. We would use the equivalent search words in Chinese databases.

Searching other resources

The following clinical trial registries will be used to retrieved ongoing trials: the NIH clinical registry ClinicalTrials.gov, the International Clinical Trials Registry Platform (ICTRP) and the Chinese clinical registry. We would retrieve the relevant SRs and meta-analyses manually and review it to identify additional studies. Useful but incomplete information would be acquired from the contact trial personnel.

Data collection and analysis

Selection of studies

According to the inclusion criteria, all retrieved studies will be assessed by reviewers

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(YFZ, YC) in the light of titles and abstracts. Full text of the qualified study will be reviewed if necessary. A third reviewer (LYH) will arbitrate if any disagreement occurs. For excluded studies, the exclusion reason will be listed. The procedure of study selection will be shown in a Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart (figure 1).

Data extraction and management

Two reviewers (YFZ, YC) will double-check and collect information from all qualifying studies and transferred into RevMan software (V.5.4).[22] A predefined data acquisition form will be entering with details: author list and affiliation, publication source, country, interventions, outcomes, adverse effects and so on. The acupuncture intervention details (such as the type of acupuncture, number of needle insertions, depths of insertion, selection of points, de-qi, needle stimulation) will be elaborated according to STandards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA).[23] A third reviewer (LYH) will arbitrate if any discrepancy noticed. We would obtain the information which is not available by contacting the authors.

Assessment of risk of bias

A systematic review of each study for the bias risk will be done by two independent reviewers (NS, YFZ) by using the Cochrane Handbook for Systematic Reviews of Interventions. Six domains which are the bias of selection, performance, detection, attrition, reporting and other sources, will be assessed. For each domain, trials are going to be rated as low risk, high risk, or unclear after evaluation.[24] Involved authors will be contacted if anything unclear. Discrepancies will be arbitrated by a third reviewer (LYH) as well.

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Measures of treatment effect

Synthesizing and statistically analyzing of the power data will use RevMan V.5.4. The dichotomous data will be analyzed using a risk ratio with 95% CI. The mean difference or standard mean difference (SMD) with 95% CI will be used to analyze the continuous data. If different evaluation tools are used, SMD will be used.

Unit of analysis issues

The analysis will be based on the aggregated outcome data, which is caused by the lack of individual patient data.

Dealing with missing data

The authors will be contacted by reviewers to obtain the missing information which was not clearly described in each study included.[25] If available, we will preferentially use the data from intention-to-treat analysis.[26] If continuous outcome data is not expressed as means and standard deviations, we will try to re-calculate means and standard deviations as the first choice.

At the same time, if feasible, we will conduct a sensitivity analyses to address the potential impact of missing data.[27] The potential impact of the effect of missing information on the final findings of the review will be addressed in the discussion.

Assessment of heterogeneity

Forest plots will be used to present effect sizes and corresponding 95% confidence limits for each study. At the same time, the inconsistency will be quantified by calculating the I^2 test.

Assessment of reporting biases

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A funnel plot for observing the reporting biases will be generated when more than ten studies are included.[25] Potential reporting biases will be further assessed with Egger's test.

Data synthesis

Data synthesis will be performed through RevMan software (V.5.4). The random-effects model will be used to pool the data. When necessary, the possible causes of the large degree of heterogeneity will be analyzed, or a subgroup analysis will be conducted. If the heterogeneity is considerable in the included trials, no meta-analysis will be performed.

Subgroup analysis and investigation of heterogeneity

Subgroup analysis will be conducted if data is available. Variations will be considered in the characteristics of the treatments for acupuncture, participants, control types. And there will be subgroups to interpret the heterogeneity.

Sensitivity analysis

The robustness of the main decisions made during the monitoring review process will be conducted through a sensitivity analysis. Several decision-making nodes for sensitivity review need to be considered in the system review process, such as methodological flaws, small research and data loss. Sensitivity analysis, as suggested in the Cochrane Handbook, includes two steps: in the first step, all major meta-analysis studies need to be included, and secondly, those studies that are known to be eligible. The results of the sensitivity analysis will be provided in the summary table. As shown by the results of the sensitivity analysis, the risk of bias will be discussed

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during the review process.

Other analysis

The use of meta-regression analysis will be considered to explore the potential heterogeneous resources.

Summary of evidence

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system approach will be used to measure the quality of main outcomes by two reviewers (NS, YFZ).[28] And the results will be present in "Summary of findings" tables. The assessments of evidence quality will be rated "high", "moderate", "low" or "very low". [29,30] Evidence of specific studies will be evaluated based on the risk of bias, inaccuracy, inconsistency, publication bias, indirectness, dose–response relation or effect size.

Patient and public involvement

No patient involved.

Ethics and dissemination

Ethics approval will not be required for this review, as it will only involve the collection of literature previously published. All included studies will be in accordance with the Declaration of Helsinki and current ethical standards. The results of this meta-analysis will be disseminated in a peer-reviewed journal or relevant conference, through publication.

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DISCUSSION

There are currently several ongoing RCTs and many others completed of acupuncture for BC. This meta-analysis will provide a relatively convincing conclusion as to whether acupuncture is effective and safe for BC. We assume the findings taken from this study would have important consequences for BC people, physicians and policymakers. If this protocol needs to be amended, we shall provide the date of each amendment and, therefore, state the reasons.

Author Contributors NS, F-RL and R-RS designed the systematic review. NS drafted the protocol，R-RS, F-RL and W-WZ revised the manuscript. NS, Y-FZ, YC, S-RC, JZ, G-XX and L-YH participated in the design of data synthesis and analysis plan. All authors have read and approved the publication of the protocol.

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Competing interests None declared.

Patient consent Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

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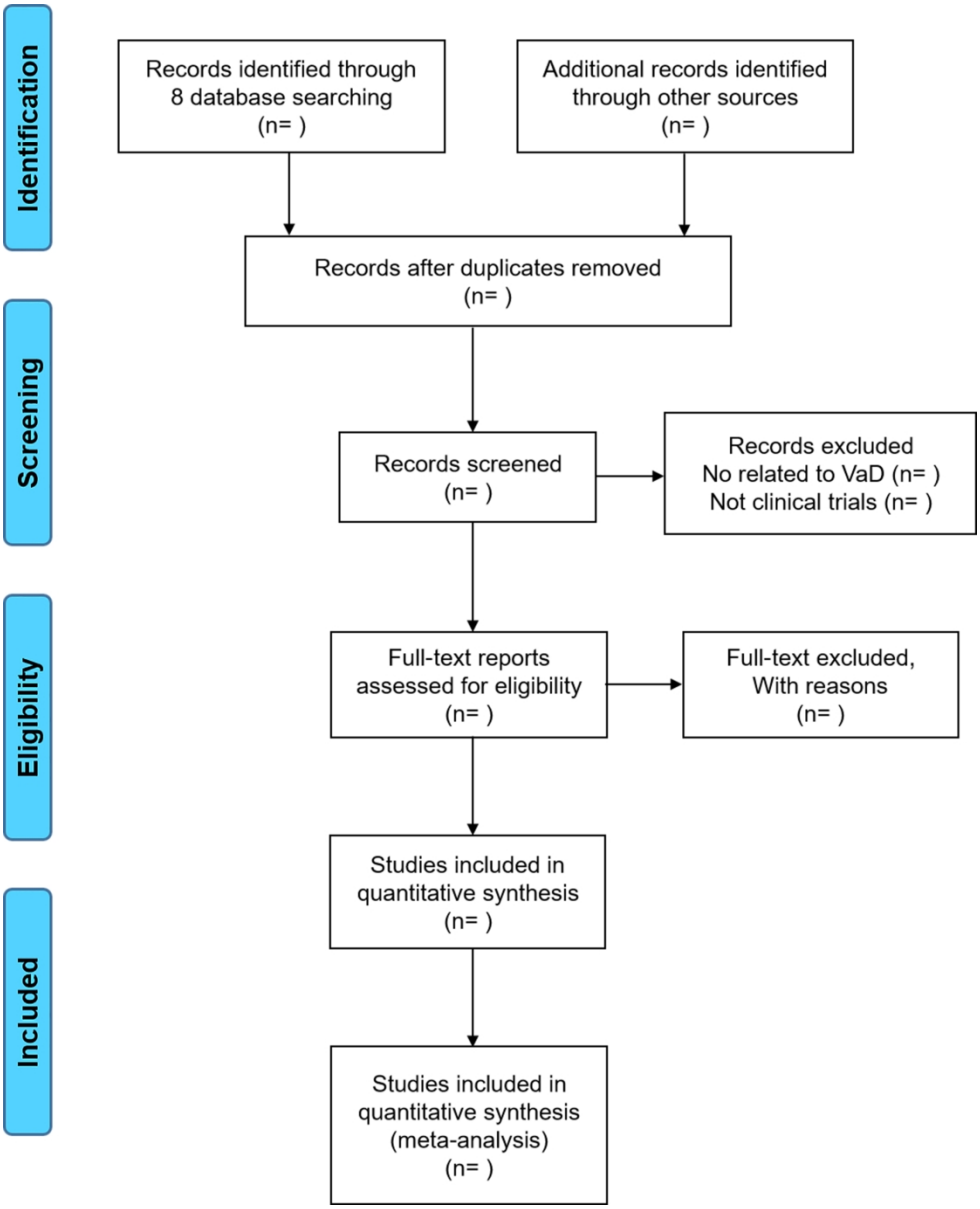
Table 1 Search strategy used in MEDLINE database

Figure 1 Flow diagram of the study selection process.

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Table 1 Search strategy used in MEDLINE database	
NO.	**Search items**
#1	randomized controlled trial [pt]
#2	controlled clinical trial [pt]
#3	randomized [tiab]
#4	placebo [tiab]
#5	clinical trials [MeSH]
#6	randomly [tiab]
#7	trial [ti]
#8	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
#9	humans [MeSH]
#10	#8 and #9
#11	biliary colic [MeSH]
#12	cholecystitis [MeSH]
#13	(biliary disease or biliary pain or biliary complication or acute cholecystitis or cholelithiasis-related complication): ti,ab
#14	#11 or #12 or #13
#15	acupuncture therapy (MeSH)
#16	acupuncture (MeSH)
#17	(acupuncture or body acupuncture or manual acupuncture or electroacupuncture or electro-acupuncture or acupoint or auricular acupuncture or warm needling): ti,ab
#18	#15 or #16 or #17
#19	#10 and #14 and #18

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Identification
Screening
Eligibility
Included
Records identified through 8 database searching (n=)
Additional records identified through other sources (n=)
Records after duplicates removed (n=)
Records screened (n=)
Records excluded No related to VaD (n=) Not clinical trials (n=)
Full-text reports assessed for eligibility (n=)
Full-text excluded, With reasons (n=)
Studies included in quantitative synthesis (n=)
Studies included in quantitative synthesis (meta-analysis) (n=)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	
Support:			
Sources	5a	Indicate sources of financial or other support for the review	12
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3-4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-5
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5-6
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be	7

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		repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7-9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	10
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	10
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	11

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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